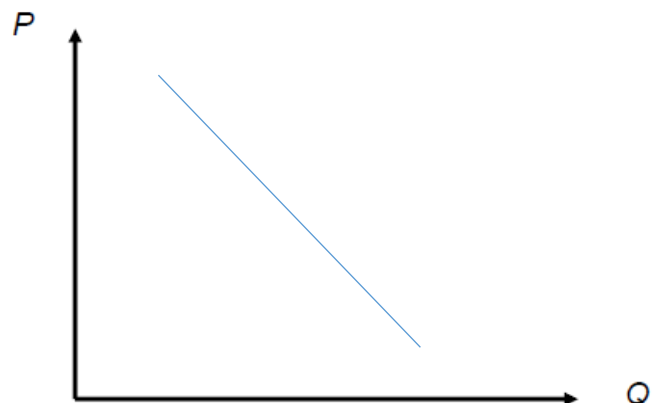


Solution to Problem set 1: Basics

A.Demand

1. Please determine for each of the following examples, whether the products are substitutes, complements or neither and explain in 1-2 sentences why.
 - a. Coffee and Tea *Substitutes*
 - b. Coffee and Sugar *Complements*.
 - c. Coffee and Cigarettes *Complements*.
 - d. Paper and Computer *Substitues for the purpose of calculating, but complements for e.g. printing.*
 - e. Houses for sale and apartments for rent. *Substitues*
 - f. Butter and butterflies *unrelated*
 - g. Gold and Bitcoin *substitutes*
2. Please circle all the correct answers (there might be more than one) for the following multiple-choice questions.
 - a. Willingness to pay is the ____ amount that ____ are willing to pay in order to ____ a given product.
 - i. Lowest, Consumers, Buy.
 - ii. Highest, Producers, Sell
 - iii. Lowest, Producers, Buy
 - iv. Highest, Consumers, Buy**
 - b. Suppose product A and product B are ____ . This means that when the price of A ____ then the demand for B _____.
 - i. Substitutes, increases, increases.**
 - ii. Complements, increases, increases.
 - iii. Substitutes, decreases, increases.
 - iv. Complements, decreases, increases.**
3. Explain in 1-2 sentences what is the “law of demand” *The law of demand is the inverse relationship between price and quantity demanded – the more expensive it is, the less people want to buy.*
4. Suppose that the following demanded quantities are given. Draw the demand function.

Price	Quantity Demanded
10	50
20	40
30	30
40	20
50	10



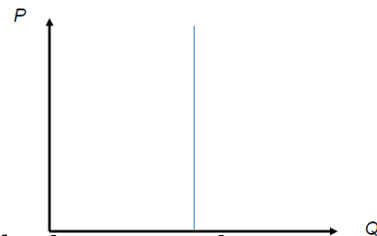
5. Suppose that the following quantities are given. Draw the demand function and determine (without calculation) the elasticity of demand. *Perfectly elastic*

Price	Quantity Demanded
10	10
10	20
10	30
10	40
10	50



6. Suppose that the following quantities are given. Draw the demand function and determine (without calculation) the elasticity of demand. *Perfectly inelastic*

Price	Quantity Demanded
10	20
20	20
30	20
40	20
50	20



7. Suppose that a hand sanitizer costs 50\$ and that for this price, each consumer is willing to buy 5 units. One day, the price increases to 70\$, and consumers are now willing to buy only 3 units each.

- a. Calculate the elasticity of demand at the point of the original price and quantity.

	Q	P
Old	5	50
New	3	70

$$Ed = \frac{p}{q} * \frac{\Delta q}{\Delta p} = \frac{50}{5} * \frac{(5 - 3)}{(50 - 70)} = 10 * \left(-\frac{1}{10}\right) = -1$$

- b. Is the demand elastic, inelastic, or unit-elastic?

The demand is unit elastic.

- c. Suppose that after the price increase, there a price decrease: each sanitizer is now sold for 40\$ and consumers are then willing to buy 6 units. Calculate the elasticity at the point of the new(est) price and quantity (i.e. after the decrease). Is the demand elastic, inelastic, or unit-elastic? (note: use the numbers given for the “old” price of 50\$; i.e. ignore the “new” price of 70\$ here).

	Q	P
After price decrease	6	40
Original	5	50

$$Ed_{after\ decrease} = \frac{p}{q} * \frac{\Delta q}{\Delta p} = \frac{40}{6} * \frac{(5 - 6)}{(50 - 40)} = \frac{20}{3} * \left(-\frac{1}{10}\right) = -\frac{2}{3}$$

The demand is relatively inelastic.

8. Suppose that a firm has the following inverse demand function:¹ $p = 10 - 2q$. Calculate the point elasticity of demand (PED) when the price equals 4.

Recall that $PED = \frac{p}{q} * \frac{\Delta q}{\Delta p}$ where the second term can also be rewritten as a (partial) derivative

$\frac{\Delta q}{\Delta p} = \frac{\partial q}{\partial p}$. We can thus follow simple steps:

1. Rewrite the demand function with q on the LHS: $q = 5 - \frac{1}{2}p$
2. Take the partial derivative w.r.t p $\frac{\partial q}{\partial p} = \frac{\partial(5 - \frac{1}{2}p)}{\partial p} = -\frac{1}{2}$
3. Insert p=4: $q = 5 - \frac{1}{2} * 4 = 3$.
4. $PED = \frac{p}{q} * \frac{\Delta q}{\Delta p} = \frac{4}{3} * \left(-\frac{1}{2}\right) = -\frac{2}{3}$

¹ An inverse demand function is expressed by the price, p, on the LHS and the quantity demanded, q, on the RHS.

B. Supply

1. Explain the difference between a fixed cost and a variable cost *Fixed costs are independent of the level of production, whereas variable costs depend on production (it is the sum of marginal production costs)*
2. Explain the difference between an average cost and a marginal cost *Average costs are (fixed costs + variable costs)/units. Marginal cost is the cost of the marginal unit, considering only variable costs.*
3. What is the production-possibility frontier (“PPF”)? *The frontier entails the combinations where one can produce the most amount of units from one product while giving up the last amount of units of the other product.*
4. Determine whether the following statements are correct or incorrect and explain why in 1-2 sentences:
 - a. If the firm produces at a point which is on the production-possibility frontier, then there is full employment. *[Technically incorrect because there may be structural unemployment]*
 - a. if the firm produces below the PPF, there is unemployment. *[Incorrect, it could be that all are employed but allocation is inefficient]*
 - b. If the price is higher than the firm’s marginal cost, then the firm should not produce. *[incorrect. it’s the opposite – if the price is lower than marginal cost, the firm shouldn’t produce]*
 - c. If the price is lower than the firm’s average cost, the firm should produce in the short run but possibly not in the long-run. *[correct, but only if the price is larger than the marginal cost]*
5. Suppose you have 100 workers: 50 workers of type A and 50 of type B. You can produce two products: Flowers and Chairs. To produce each product, 25 workers are needed. The output table is:

	Flowers	Chairs
Worker A	1	5
Worker B	2	8

Which type of worker should be allocated for producing which product?

Solution: Opportunity cost table

To produce:	1 Flower	1 Chair
Worker A gives up	5 Chairs	1/5 flower
Worker B gives up	4 chairs	1/4 flower

Thus, Worker A should produce chairs. Worker B should produce flowers.

6. Suppose that the supply curve's formula is $p = 12 + 3q$.
 - a. Explain why there is a 'plus' sign (rather than a 'minus' sign) on the RHS. *the plus sign captures the higher willingness to produce when the price is higher.*
 - b. If the price in the market is 42, how many units are firms willing to produce?
 $42 = 12 + 3q \rightarrow q = 10$.

C. Equilibrium

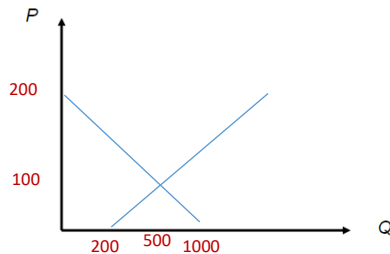
1. The demand and supply functions for Coffee are given by:

$$Q^d = 1000 - 5p$$

$$Q^s = 200 + 3p$$

Where Q^d is the quantity demanded and Q^s is the quantity supplied.

- a. Plot the supply and demand curves on a graph and show where the equilibrium occurs.



- b. Calculate the equilibrium price and quantity (using algebra).

In Equilibrium, quantity demanded is equal to quantity supplied. Hence,

$$Q^d = Q^s \Leftrightarrow 1000 - 5p = 200 + 3p$$

$$p^* = 100, q^* = 500$$

2. A manufacturer in a perfectly competitive market produces product X using one input factor:
I. Suppose that price of *I* increases but nothing else changes. The manufacturer will then:
- (1) Increase the produced quantity
 - (2) Raise the price of product X
 - (3) Decrease the produced quantity**
 - (4) Answers (1) and (2) are both correct
 - (5) Answers (1) and (3) are both correct

3. Here are two claims about a firm operating in a perfectly competitive market:
- (1) If the variable cost for producing 3 units is 300\$, the average cost per unit for producing 4 units is 100\$ and the firm has a fixed cost of 80\$, then the marginal cost is 20\$. *Correct*

<i>Units produced</i>	<i>Fixed cost</i>	<i>Average cost</i>	<i>Total costs</i>	<i>Total variable cost</i>	<i>Marginal cost</i>
3	80			300	
4	80	100	400	400-80=320	20

- (2) The firm has an increasing marginal cost function and produces 20 units. If the firm increase the price in which it sell a unit by 3\$, the profits of the firm will increase by 60\$.

Claim 2 is incorrect, for two reasons. First, the firm's marginal costs will increase when it produces more units. Second, it's a perfectly competitive market, so If the firm unilaterally increases the price it will actually sell nothing.

Are these claims correct? If not, explain why not.

4. A firm is a monopoly. Therefore:
- (1) **It has a downward-sloping marginal revenue curve.**
 - (2) It has an upward-sloping marginal revenue curve
 - (3) It has a fixed marginal revenue.
5. A firm is a monopoly. Therefore its marginal revenue curve is:
- (1) **Below the demand curve.**
 - (2) Above the demand curve.
 - (3) Identical to the demand curve.
6. Suppose that the price of coffee beans increases and the demand for Capuccino increases. What will happen to the price of a cup of Capuccino?
- (1) **The price will increase**
 - (2) The price will decrease

- (3) The price will not change
 - (4) There is not enough information to determine whether the price will increase or decrease.
7. Suppose that the price of tea increases and the supply of coffee beans increases. What will happen to the price of a cup of Capuccino?
- (1) The price will increase
 - (2) The price will decrease
 - (3) The price will not change
 - (4) **There is not enough information to determine whether the price will increase or decrease.** *[Tea is a substitute; thus a price increase implies an increase in demand for coffee. But there is also an increase in supply. So we cannot determine]*
8. Suppose that the price of a keyboard increases and the supply of computers increases. What will happen to the price of a computer?
- (1) The price will increase
 - (2) **The price will decrease** *[keyboard is a complementary product. Its price increase will reduce demand of computers. As there is also an increase in supply, the price decreases]*
 - (3) The price will not change
 - (4) We cannot determine whether the price will increase or decrease.
9. Suppose that the demand for antibiotics is completely inelastic. A new and cheaper way to produce antibiotics is discovered. What will happen to the price of antibiotics? What will happen to the quantity sold?
- (1) The price will increase and the quantity will decrease.
 - (2) The price will increase and the quantity will increase.
 - (3) The price will not change but the quantity will increase.
 - (4) **The price will decrease but the quantity will not change.**
10. Suppose that the demand for cola is completely elastic. The wage of workers decreases. What will happen to the price of cola? What will happen to the quantity sold?
- (1) The price will increase and the quantity will decrease.
 - (2) The price will increase and the quantity will increase.
 - (3) **The price will not change but the quantity will increase.**
 - (4) The price will decrease but the quantity will not change.

D.Efficiency - basics

1. Explain the difference between Pareto efficiency and Kaldor-Hicks efficiency. *The definitions appear in the course's slides.*
2. Which of the following distributions of money between "A" and "B" is Pareto *inefficient*? Which distribution is Kaldor-Hicks *efficient*?

Distribution 1: A has 20. B has 20.

Distribution 2: A has 10. B has 20.

Distribution 3: A has 15. B has 25

Distribution 4: A has 100. B has 5.

Distribution 2 is Pareto inefficient (moving to (1) to (3) is a pareto improvement). All the rest are Pareto efficient. Distribution 4 is KH efficient.

3. I have A car, but I like scooters more. You have a scooter, but you like cars more. If we exchange my car for your scooter, this is...
 - (1) Inefficient
 - (2) **Pareto efficient**
 - (3) **Kaldor-Hicks efficient**
 - (4) Not enough information to determine
4. We have the choice between 4 distributions:
 - (1) I get 5\$, you get 5\$
 - (2) I get 3\$, you get 10\$.
 - (3) I get 15\$, you get 2\$.
 - (4) I get 10\$, you get 7\$

Which distribution is pareto inefficient? Which distribution is Kaldor-hicks efficient?
Option (1) is pareto inefficient (dominated by (4)). Options 3 and 4 are KH efficient.

5. We have the choice between 4 distributions:
 - (1) I get 1\$, you get 1\$
 - (2) I get 2\$, you get 7\$.
 - (3) I get 9\$, you get 2\$.
 - (4) I get 10\$, you get 10\$

Which distributions are pareto efficient? Which distribution is Kaldor-hicks efficient? **Only option 4 is Pareto efficient (as it gives more to both compared to other option). Option 4 is also the only KH-efficient option, as it maximizes the joint amount (10+10=20).**

6. We have the choice between 3 distributions:

- (1) A gets 3\$, B gets 8\$.
- (2) A gets 100\$, B gets 2\$,
- (3) A gets 4\$ B gets 4\$

Which distribution is optimal according to Rawls? *Option 3 (the poorest has the most: $4 > 3 > 2$).*

7. A firm's CEO promises not to embezzle money, but the board of directors fears he might do so anyway. The firm is trying to decide whether to create an algorithm that automatically discloses embezzlements to the police. Discuss what is the tradeoff that the firm faces from an ex-ante and an ex-post efficiency perspective.

From an ex-ante perspective, the algorithm might deter the CEO from stealing, which is efficient. But from an ex-post perspective, disclosing the police will cause a loss of reputation and has little benefit once the money was stolen.

E.Surplus

- 1. Draw a set of standard supply and demand curves and mark the consumer surplus and the supplier surplus. *See the slides.*
- 2. What is the consumer surplus when the demand is (1) perfectly elastic and (2) perfectly inelastic? *If demand is perfectly inelastic, the CS is infinity. If it is perfectly elastic, the CS=0.*
- 3. Explain why the supplier surplus increases as the supply becomes inelastic. *Because the seller is willing to sell for any price so that there is more benefit from selling.*